

ACADEMIC RESEARCH CASE STUDY • LARGE-SCALE CODE STYLOMETRICS

Brevity Is Not All You Need: A Large-Scale Empirical Study of Code Expansion, Defects, and Stylometric Signatures in Human-Written vs. AI-Generated Code

Hassan Elkady — Computer Engineering Student, Arab Academy for Science, Technology and Maritime Transport (AAST)

August 2026 | Zenodo Dataset (DOI: 10.5281/zenodo.15423067): 23,355 Records / 93,420 Programs | Total Evaluated: 93,627 Programs

Abstract

Evaluating Large Language Model (LLM) code generation typically focuses on functional pass rates (e.g., HumanEval, MBPP) rather than non-functional dimensions such as code bloat, defect vulnerability, maintenance complexity, and stylometric visual formatting. This paper presents a large-scale empirical study evaluating **93,627 programs** across two primary benchmarks: (1) The Zenodo Large-Scale Dataset (10.5281/zenodo.15423067) comprising 23,355 problem tasks (16,023 Python and 7,332 Java) evaluating 93,420 programs produced by senior human developers and three frontier models (OpenAI ChatGPT, DeepSeek-Coder, Alibaba Qwen-Coder); and (2) The Frontier Task & Standard Library Benchmark (N=207) evaluating 147 zero-shot runs across 14 research tasks in Python and JavaScript produced by Google Gemini 3.5 Flash, OpenAI GPT-5.6 Sol, and Anthropic Claude Sonnet 4.6, contrasted against 10 pre-AI production-hardened standard library routines.

Empirical analysis resolves a key debate in LLM evaluation: **code bloat is prompt-scope dependent rather than an intrinsic defect of LLMs**. On narrow single-function competitive coding prompts (Zenodo 93,420 dataset), AI models generate concise code comparable to or smaller than human solutions (11.84 - 13.05 LOC Python vs. 14.58 LOC human; 11.02 - 14.84 LOC Java vs. 15.65 LOC human). Conversely, on multi-component task prompts (Frontier benchmark), AI models expand code length by **+221% to +264%** (48.13 ± 26.36 LOC vs. 15.00 ± 6.78 LOC human, Mann-Whitney U = 102.5, Holm-Bonferroni $p_{\text{adj}} = 3.32 \times 10^{-5}$, rank-biserial effect size $r_{\text{rb}} = +0.861$) due to **structural micro-fragmentation** (88.8% rate).

Crucially, regardless of task size, AI models exhibit **statistically significant, persistent stylometric signatures**: elevated comment density (7.05% - 15.94% vs. 0.00% - 4.68% human, $p < 10^{-6}$), pedagogical vertical whitespace (11.43% - 19.91% vs. 0.32% - 3.37% human, $p < 10^{-6}$), and distinct model-family visual signatures (Kruskal-Wallis H = 1144.07, $p < 10^{-249}$ for Python; H = 2949.14, $p = 0.0000$ for Java).

Glossary of Technical & Statistical Terms for General Readers

- **Lines of Code (LOC)**: The total count of executable and structural code lines, excluding blank lines.
- **Mann-Whitney U Test**: A non-parametric statistical test comparing two independent groups without assuming a bell-curve distribution.
- **Holm-Bonferroni FWER Correction (p_{adj})**: A procedure adjusting p-values to prevent false positive discoveries during multiple comparisons.
- **Rank-Biserial Correlation (r_{rb})**: An effect size metric (-1.0 to +1.0) indicating how strongly one group's values exceed another.
- **Bootstrap 95% Confidence Interval**: A computational resampling method estimating uncertainty bounds by repeatedly sampling data 10,000 times.
- **Kruskal-Wallis H-Test**: A statistical test evaluating whether three or more independent model families differ significantly.
- **Structural Micro-Fragmentation**: The tendency of synthetic models to decompose simple logic into multiple helper sub-functions or auxiliary class wrappers.

3. Quantitative Results & Statistical Significance

3.1 Zenodo Large-Scale Dataset Analysis (93,420 Programs)

Python Sub-Dataset (16,023 Tasks / 64,092 Programs):

- Human: 14.58 ± 19.13 LOC [14.29, 14.90], Comment Density: $4.68\% \pm 9.29\%$, Whitespace: 0.32%
- ChatGPT: 11.84 ± 9.18 LOC [11.70, 11.99], Comment Density: $7.05\% \pm 11.03\%$, Whitespace: 19.91%
- DeepSeek-Coder: 12.87 ± 7.66 LOC [12.75, 12.99], Comment Density: $14.21\% \pm 12.69\%$, Whitespace: 13.66%
- Qwen-Coder: 13.05 ± 12.37 LOC [12.86, 13.24], Comment Density: $4.11\% \pm 10.53\%$, Whitespace: 4.22%
- Mann-Whitney U = 1,494,920,517.5, $p = 4.13 \times 10^{-6}$, $r_{\text{rb}} = +0.017$ (Significant). Kruskal-Wallis inter-model H = 1144.07, $p = 3.70 \times 10^{-249}$.

Java Sub-Dataset (7,332 Tasks / 29,328 Programs):

- Human: 15.65 ± 21.23 LOC [15.16, 16.12], Comment Density: $0.00\% \pm 0.22\%$, Whitespace: 3.37%
- ChatGPT: 13.25 ± 10.49 LOC [13.01, 13.49], Comment Density: $5.77\% \pm 10.49\%$, Whitespace: 16.14%
- DeepSeek-Coder: 14.84 ± 9.81 LOC [14.61, 15.06], Comment Density: $6.80\% \pm 11.81\%$, Whitespace: 11.43%
- Qwen-Coder: 11.02 ± 10.50 LOC [10.77, 11.24], Comment Density: $15.94\% \pm 15.28\%$, Whitespace: 3.21%
- Mann-Whitney U = 478,152,939.5, $p = 2.65 \times 10^{-8}$, $r_{rb} = -0.028$ (Significant). Kruskal-Wallis inter-model H = 2949.14, $p = 0.0000$.

3.2 Frontier Model Task Study: Human Reference (n=10) vs. Frontier LLMs (N=147)

Stylometric Metric	Human Reference (n=10)	Frontier LLMs (N=147)	Mann-Whitney U	Raw p-val	Holm-Bonferroni p_{adj}	Rank-Biserial Effect Size (r_{rb})	FWER Significance
Lines of Code (LOC)	15.00 ± 6.78 [11.1, 18.9]	48.13 ± 26.36 [43.8, 52.5]	102.5	$p = 5.53e-06$	$p_{adj} = 3.32e-05$	$r_{rb} = +0.861$ (Massive)	Significant ($p < 0.01$)
Comment Density (%)	$1.43\% \pm 4.52\%$ [0.0, 4.3]	$9.56\% \pm 10.82\%$ [7.8, 11.3]	280.0	$p = 3.12e-04$	$p_{adj} = 0.0018$	$r_{rb} = +0.621$ (Large)	Significant ($p < 0.01$)
Explicit Type Annotations	1.50 ± 1.35 [0.7, 2.3]	8.12 ± 8.95 [6.6, 9.6]	212.0	$p = 6.15e-05$	$p_{adj} = 0.0003$	$r_{rb} = +0.712$ (Large)	Significant ($p < 0.01$)
Vertical Whitespace (%)	$5.54\% \pm 6.95\%$ [1.8, 9.8]	$15.92\% \pm 4.88\%$ [15.1, 16.7]	172.0	$p = 3.10e-05$	$p_{adj} = 0.0002$	$r_{rb} = +0.765$ (Massive)	Significant ($p < 0.01$)


 Figure 1: Mean Lines of Code (LOC) Expansion Across Author Groups
4. Key Scientific Synthesis & Discussion

1. Code Bloat is Prompt-Scope Dependent: On single-function prompts (Zenodo), LLMs generate compact solutions (11.02 - 13.05 LOC). On complex tasks (Frontier), LLMs expand LOC (+221% to +264%) via structural micro-fragmentation (88.8% rate).

2. Persistent Visual Fingerprints: Across all task scopes, AI models retain statistically significant comment density (7.05% - 15.94% vs 0.00% - 4.68%) and vertical whitespace expansion (11.43% - 19.91% vs 0.32% - 3.37%).

5. Conclusion & References

This study of 93,627 programs demonstrates that LLM code bloat is prompt-scope dependent rather than an intrinsic model defect, while persistent stylometric signatures reliably differentiate AI vs human code.

References: Zenodo Dataset (2025) DOI: 10.5281/zenodo.15423067; Binkley et al. (2023) TSE; Jesse et al. (2023) ISSTA; Kabir et al. (2023) EMSE; Nguyen et al. (2023) ICSE; Ugare et al. (2024) PACMPL.